Machine Learning and Data Science Concepts


Author: Landry CYIZA MPAYIMANA
ETEFOP TSS

CONTENTS

1. SECTION A: Multiple Choice (20%)

2. SECTION B: Matching (10%)

3. SECTION C: Fill in the Blanks (10%)

4. SECTION D: Open Questions (50%)

01

SECTION A: Multiple Choice (20%)

Questions and Answers

No	Question	Correct Answer
1	What is Machine Learning?	b) A part of AI that learns from data
2	First step in ML life cycle?	b) Problem definition
3	Tool for data cleaning in Python?	b) Pandas
4	Data with labels is used in?	c) Supervised learning
5	6 V’s of Big Data?	b) Volume, Variety, Velocity, Veracity, Variability, Value
6	Used to check model accuracy?	b) Precision & Recall
7	Cloud IDE for ML with GPU/TPU?	c) Google Colab
8	Algorithm for clustering?	a) K-Means
9	Disadvantage of ML?	b) High cost for computation
10	Built-in dataset in Scikit-learn?	c) Iris dataset

02

SECTION B: Matching (10%)

Type and Meaning

Type	Match	Meaning
1. Supervised Learning	D	Uses labeled data to make predictions
2. Unsupervised Learning	A	Groups unlabeled data
3. Reinforcement Learning	B	Learns from trying and getting rewards
4. Semi-supervised Learning	C	Uses both labeled and unlabeled data

03

SECTION C: Fill in the Blanks (10%)



SECTION C: Fill in the Blanks (10%)

Most used language in ML is Python.

NumPy is used for math and numbers.

In supervised learning, data must be labeled.

A data warehouse stores data from many sources.

Google’s ML framework is TensorFlow.

04

SECTION D: Open Questions (50%)



Questions and Answers

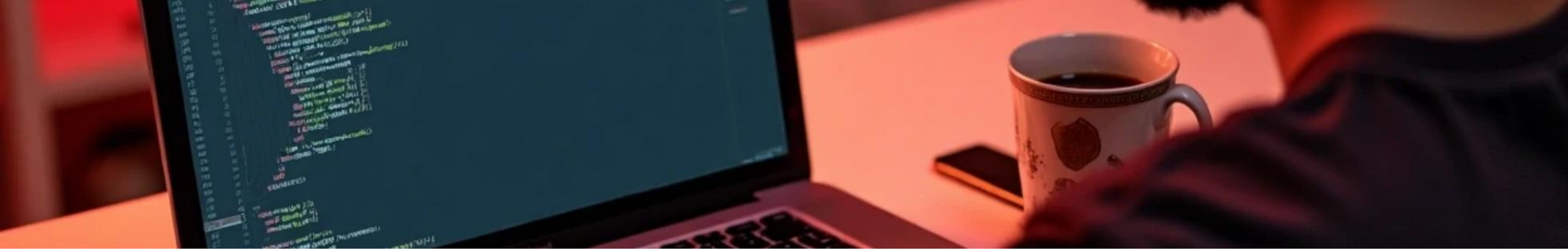
Difference between AI, ML, and DL

AI: Makes machines act like humans.

ML: Makes machines learn from data.

DL: Uses many layers of neural networks.

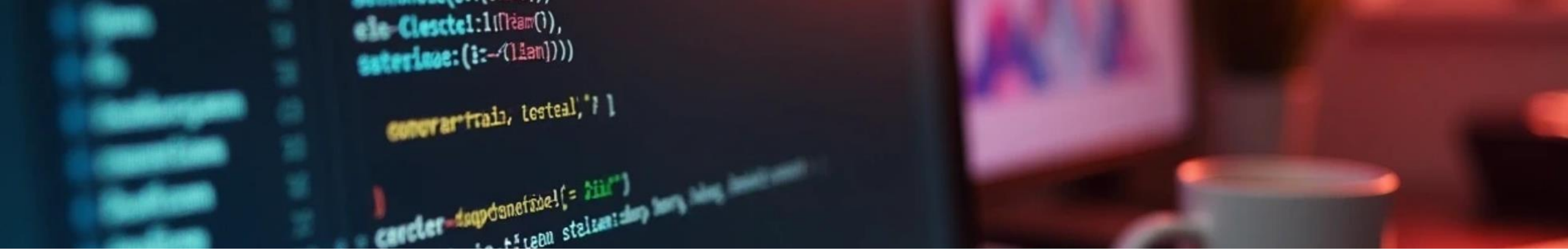
AI is the big idea, ML is a part of AI, and DL is a part of ML.



Questions and Answers

Installing Python

1. Go to python.org.
2. Download and install Python.
3. Check “Add Python to PATH.”
4. Open CMD → type python --version to check.



Questions and Answers

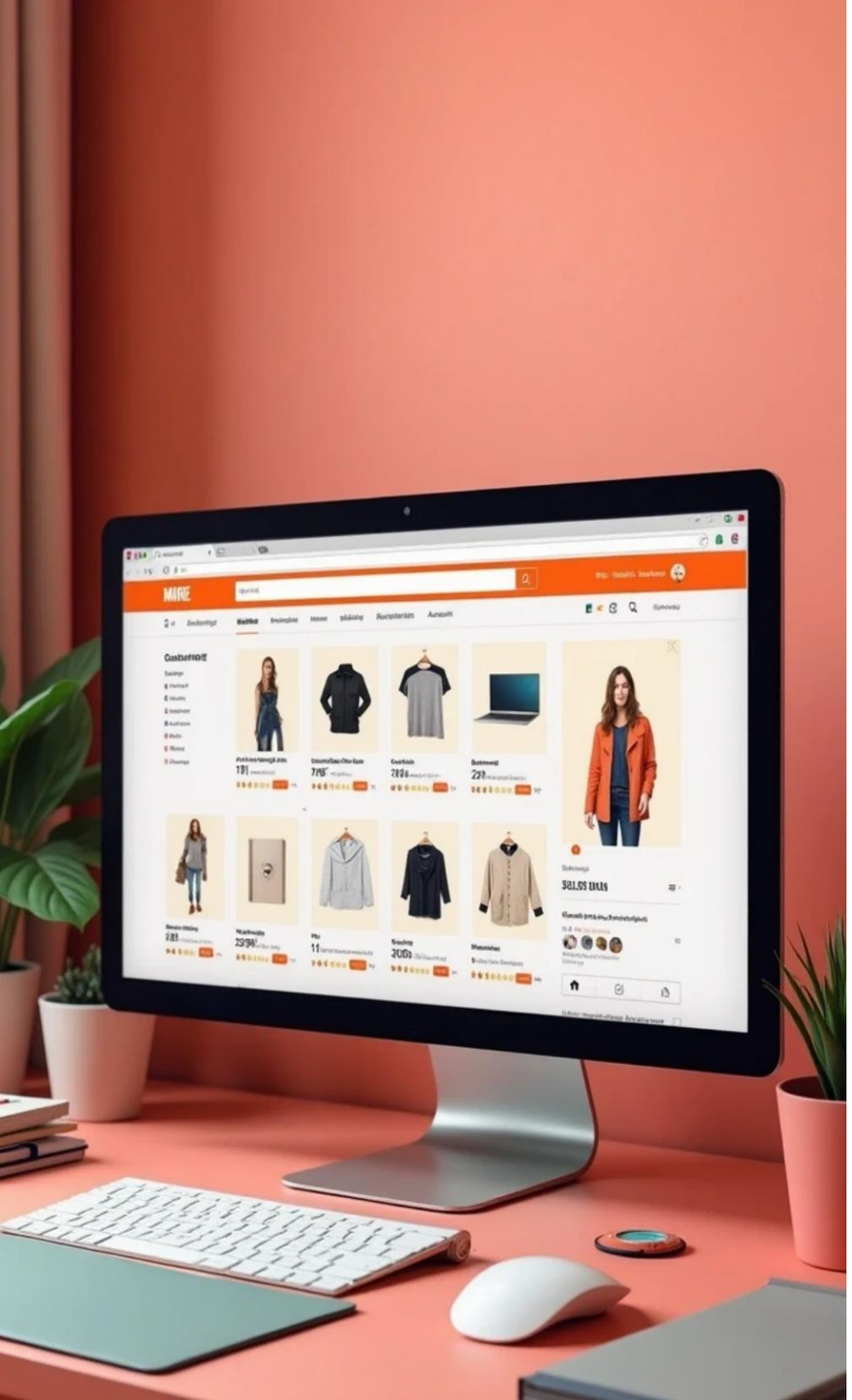
Cleaning Data

Steps

Remove missing data, remove duplicates, fix errors.

Code

import pandas as pd df = pd.read_csv('data.csv') df.drop_duplicates(inplace=True)



Questions and Answers

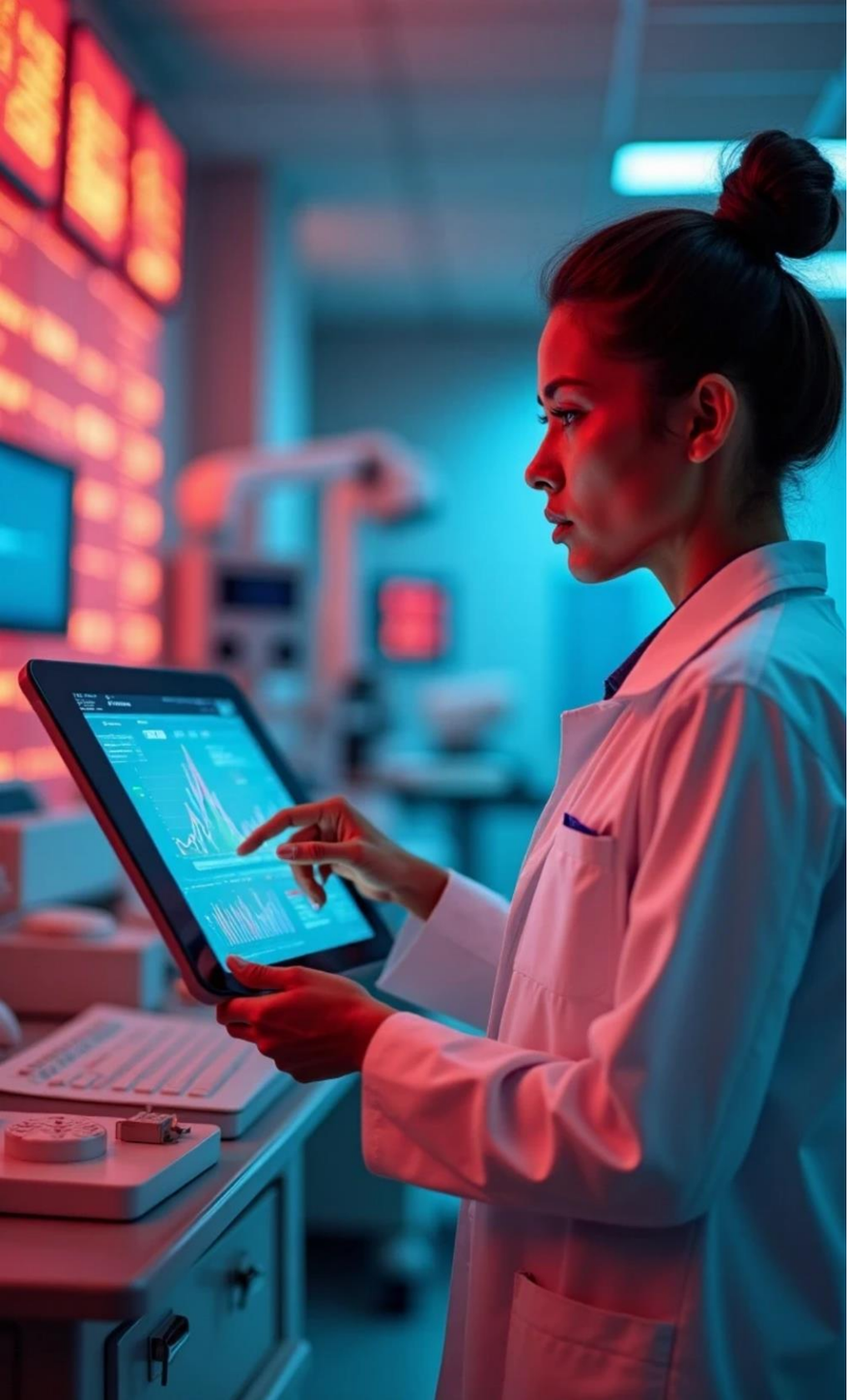
Online Store Recommendation

1

Use Machine Learning because it learns from customer data.

2

Algorithm: K-Nearest Neighbors (KNN) or Collaborative Filtering.



Questions and Answers

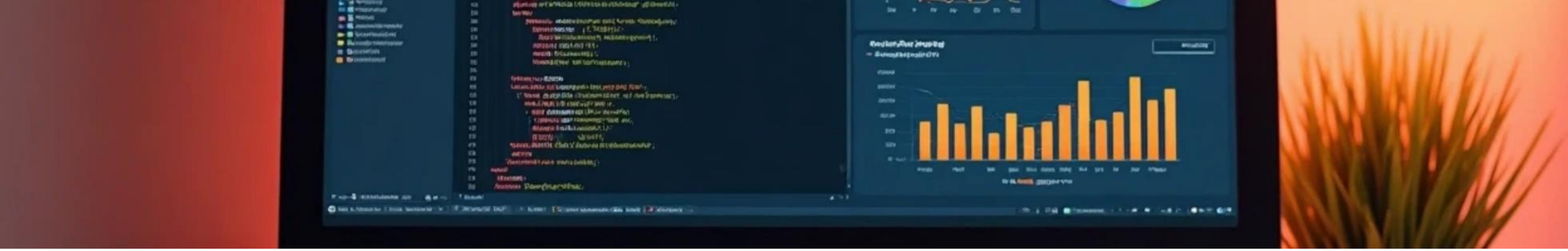
Big Data in Healthcare

✓ Advantages

- Helps doctors make better decisions.
- Can predict diseases early.

✕ Disadvantages

- Expensive to manage.
- May cause data privacy problems.



Questions and Answers

Mini Project Example: Predict Student Marks

1. Collect student data (marks, attendance).
2. Clean and prepare the data.
3. Train the model using Python.
4. Show results with graphs.

Thank You